ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION **SUMMER SEMESTER, 2022-2023**
DURATION: 1 HOUR 30 MINUTES **FULL MARKS: 75**

CSE 4849: Human Computer Interaction

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Suppose you are tasked to design and develop a workplace well-being and stress management App due to the evolving dynamics of modern work environments, where professionals navigate stressors and challenges. The App offers tangible solutions for personalized stress assessment, adaptive coping, and fostering a supportive community. In this context, concerning the interrelated Human-Computer Interaction (HCI) aspects, answer the following:

   i. How can the app practically enhance social connectivity among professionals, fostering virtual community building and peer support? 5 (CO1) (PO1)

   ii. In practical terms, how does the app align with human information processing, especially in providing personalized stress assessments that consider cognitive processes? 5 (CO1) (PO1)

   iii. Provide examples illustrating how the app practically interacts with various input/output devices, ensuring a seamless user experience across different devices in real-life scenarios. 4 (CO1) (PO1)

   iv. What design approaches are practically employed in the app's development to ensure that it is user-centered and caters to the needs of professionals? 5 (CO1) (PO1)

   b) Consider a captcha-based user login interface where a 4×4 grid of images is presented to the user. The task is to find and recognize three specific objects, such as motorcycles, within the grid of 16 images. Once the user identifies these objects, s/he is required to click the "Submit" button. The Model Human Processor (MHP) constants are given as, perceptual cycle time (PCT) is 100 ms, cognitive processing time (CPT) is 70 ms, and motor processing time (MPT) is 120 ms. Calculate the total time required for the user to complete the task using MHP. 6 (CO1) (PO1)

2. a) In a graphical user interface, a pointing task involves four targets labeled as A, B, C, and D, placed at varying distances from the starting point. The distances (D) from the starting point to the center of each target are as follows: $D_A = 120$ pixels, $D_B = 160$ pixels, $D_C = 100$ pixels, and $D_D = 140$ pixels, The widths (W) of the target areas are: $W_A = 30$ pixels, $W_B = 20$ pixels, $W_C = 25$ pixels, and $W_D = 15$ pixels. Constants for movement time (MT) calculations are, $a = 100$ ms, $b = 50$ ms. Calculate human throughput using Fitts' law for each target and determine which target provides the highest throughput. 9 (CO1) (PO1)

   b) In a collaborative virtual design platform, architects and interior designers engage in shared virtual reality (VR) sessions to collaboratively work on a detailed 3D model of an office space. Users wear VR headsets for an immersive experience, aiming to enhance the collaborative design process. Identify the physiological and psychological depth cues relevant to achieving the following tasks and explain within 1-2 lines of justification for each cue.

   i. Architects and interior designers need to precisely adjust the placement of furniture items within the virtual office space with varied distances between objects. Users manipulate objects using hand controllers to achieve accurate positioning. 4 (CO2) (PO2)

ii. The virtual environment includes textured walls with visible bricks and wood grains. Users navigate the space and assess design elements in an occluded environment. 4 (CO2) (PO1)

iii. The VR environment simulates realistic lighting conditions, with sunlight streaming through windows and casting shadows across the floor. Users can adjust the time of day to observe changing light patterns. 4 (CO2) (PO1)

iv. Users have the option to adjust the stereo separation settings in their VR headsets to optimize depth perception. This customization allows users to find the most comfortable viewing experience. 4 (CO2) (PO1)

3. a) Suppose you want to design a command line application for a certain task. The application has up to 30 configurable parameters. In your current design, the user may pass the value of each of the arguments or parameters while calling it from the terminal. Though `help` can be invoked to list down all the possible parameters, novice users still find it difficult to properly use the application. In many cases, for a certain user or environment, a lot of the parameters remain the same. For simplicity of understanding, the invocation of a command line tool, `gcc` for compiling a single C file is:

```
gcc filename.c -o Application
```

The program takes 3 arguments, the first one being the file name to compile, then `-o` to indicate that the next argument would be the output file name/path and then accordingly, the output file name. Now based on this scenario, answer the following questions.

i. Point out how human memory, both short-term and long-term, can affect the usability of the aforementioned application. 5 (CO2) (PO1)

ii. Propose an ideal solution (a single one based on your decision) to the problem of having 30 parameters in the aforementioned program. You cannot decrease the number of parameters 5 (CO2) (PO3)

b) Suppose you are hired as an interaction designer to evaluate user interactions in a newly developed mobile banking service application. This application allows users to perform financial transactions, check account balances, and manage their accounts using touch gestures on their smartphones. Users input commands through the app, and the system provides feedback on the execution of these commands. Based on the Abowd and Beale interaction framework, identify and explain the interaction problems (articulation, performance, presentation, and observation) that may exist (one or more) for the following tasks within 1-2 line justification for each problem you found.

i. Users attempt to transfer money to another account using the app. Some users encounter difficulties in accurately inputting recipient details, leading to errors in the transaction, and there are delays in the system's response to transfer requests. 5 (CO3) (PO1)

ii. Users navigate through the app to check their account statements. Some users express confusion with the presentation of transaction details, finding it difficult to interpret the information regarding deposits, withdrawals, and balances. 5 (CO3) (PO1)

iii. Users attempt to change their account PIN using the app. However, there are reports of users struggling to observe the confirmation prompts for the new PIN, leading to uncertainty about the success of the PIN change process. 5 (CO3) (PO1)

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Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION SUMMER SEMESTER, 2022-2023

DURATION: 1 HOUR 30 MINUTES FULL MARKS: 75

CSE 4803: Graph Theory

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Consider the maze in Figure 1 where the numbers represent the corridors on each side of the maze. Suppose, you start from *1* and are supposed to find your way to the center labeled as *'Destination'*. 7×5 (CO1) (PO1)

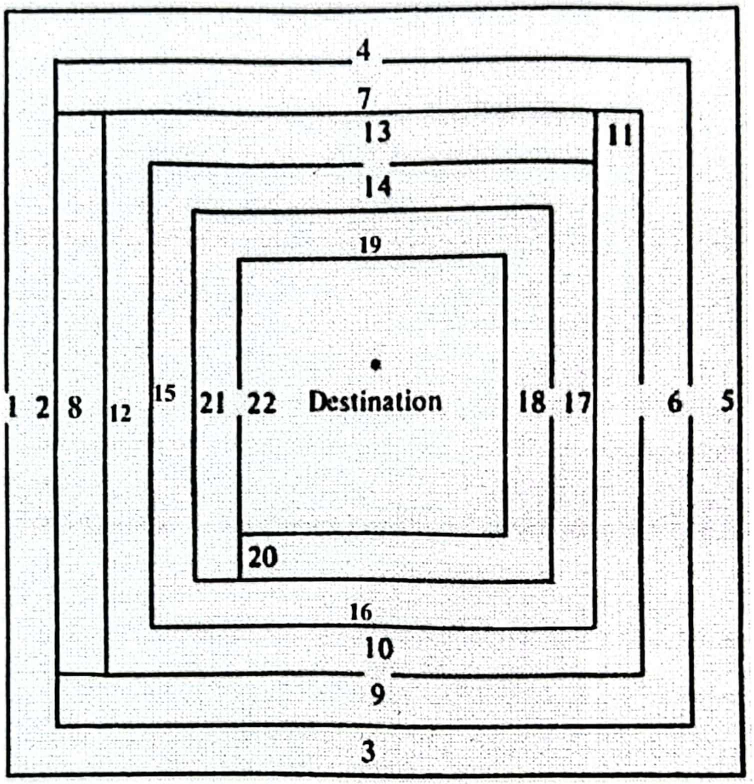

Figure 1: A drawing of a maze for Question 1

a) Formulate a graph to appropriately represent the maze shown in Figure 1.

b) Determine whether the graph from Question 1.a has a Unicursal line.

c) Justify whether the graph from Question 1.a is an Euler graph. If it is not, how can you make it Euler?

d) Determine whether the graph from Question 1.a has a Hamiltonian path.

e) Find the center(s) of the graph from Question 1.a.

f) How many edges must be removed to make the graph from Question 1.a a spanning tree? Draw one such spanning tree.

g) Considering the spanning tree from Question 1.f, draw one fundamental circuit and one fundamental cut-set.

2. Answer the following questions: 2×10 (CO1) (PO1)

   a) Prove that there can be no path longer than a Hamiltonian path (if it exists) in a graph.

   b) Prove that if a connected graph is arbitrarily traceable, it is an Euler graph.

3. Answer the following questions: 4×5 (CO1) (PO1)

   a) Show that, in a group of seven people, it is impossible for every person to be friends with exactly three other people.

   b) Justify whether a connected graph with more than six odd-degree vertices can be decomposed into only three paths.

   c) Is a vertex v of a graph G a cut-vertex if G is arbitrarily traceable on v?

   d) Prove that every vertex in a circuit has degree 2.

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MID SEMESTER EXAMINATION | SUMMER SEMESTER, 2022-2023

DURATION: 1 HOUR 30 MINUTES | FULL MARKS: 75

CSE 4801: Compiler Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Draw the block diagram of a standard compiler showing various modules with input-output. Discuss the functions of any three modules from the diagram. 10 (CO1) (PO1)

   b) The communication between lexical analyzer and syntax analyzer can be implemented by **batch** process or **on-demand** process. Explain both of the processes. 10 (CO1) (PO1)

   c) Input-buffering is a technique used in compilers to read input characters from a source file and store them in a buffer before parsing them. Explain *buffer-pair* technique to implement input-buffering. 5 (CO1) (PO1)

2. a) Discuss about various syntax errors and their recovery strategies. 9 (CO2) (PO1)

   b) Which type of parsing method can not process left recursive grammar? Explain briefly. 3 (CO2) (PO2)

   c) Predictive parsing is a special type of top-down parsing where no backtracking is required. Show the construction steps of a predictive parse table for the grammar shown in Table 1. 13 (CO2) (PO1)

Table 1: A context free grammar for Question 2.c

$$G \rightarrow G + H$$
$$G \rightarrow H$$
$$H \rightarrow H * I$$
$$H \rightarrow I$$
$$I \rightarrow \mathbf{id}$$
$$I \rightarrow (G)$$

3. a) Write short notes on *handle pruning*. 5 (CO2) (PO1)

   b) Construct a shift-reduce parser for the grammar shown in Table 2. First, find the LR(0) collection of items for the grammar and then construct the parse table. 12 (CO2) (PO1)

Table 2: A context free grammar for Question 3.b

$$S \rightarrow AA$$
$$A \rightarrow aA$$
$$A \rightarrow b$$

c) Code Snippet 1 shows a list of predefined *Lex* variables and functions. Describe their functionalities. 8 (CO1) (PO1)

```
int yylex(void)
char *yytext
yyleng
yylval
int yywrap(void)
FILE *yyout
FILE *yyin
ECHO
```

Code Snippet 1: A list of Lex predefined variables and functions for Question 3.c

B.Sc. Engg. CSE 8^{th} Semester — 06 March 2024 (Morning)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION — **SUMMER SEMESTER, 2022-2023**

DURATION: 1 HOUR 30 MINUTES — **FULL MARKS: 50**

CSE 4809: Algorithm Engineering

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Answer the following questions in brief: — 2×4 (CO2) (PO1)

   i. Why do we look for asymptotic bound of any algorithm?

   ii. Mention the characteristic(s) of algorithms that have a complexity of $lg(n)$.

   iii. How does divide and conquer technique help merge sort algorithm?

   iv. Why does quick sort algorithm just have division (i.e. partition) but does not have any merging?

   b) Write down the rationale of master method for its all three cases for determining the complexity of recursive algorithms. — 5 (CO1) (PO1)

   c) Given a regular recursion $T(n) = aT(n/b) + f(n)$
   If $f(n) = O(n^{\log_b a - \epsilon})$, prove that $T(n) = (n^{\log_b a})$. — 5 (CO1) (PO1)

2. a) Answer the following question in a single sentence: — 1×5 (CO2) (PO1)

   i. How does dynamic programming save computation of a combinatorial optimization problem?

   ii. Can dynamic programming algorithm be used in a path finding problem where the problem is to find the list of paths within a ratio of the optimal paths?

   iii. What is Catalan number? How is it related to optimal parenthesis problem?

   iv. What is optimal substructure equation?

   v. Why longest simple path finding problem does not have an optimal substructure property?

   b) Write down the optimal substructure equation for Dynamic Time Warping (DTW) after briefly describing the optimization problem the algorithm attempts to solve. Comment on the complexity of the DTW algorithm. Briefly describe two ways to improve the time complexity of the algorithm. — 6 (CO2) (PO1)

   c) Given a grid of $(m \times n)$ dimension containing cells filled with reward (positive or negative), the reward of an area will be sum of rewards of all the cells withing the area. — 5 (CO2) (PO2)
   If a problem is defined as to find the maximum possible reward from a minimum square:

   i. Can you use Dynamic Programming algorithm for the problem defined above? If yes, what will be the optimal substructure for the problem?

   ii. If Dynamic Programming cannot be used or the benefit of using the algorithm is hindered for some reason, explain why it might be the case.

3. a) Write an algorithm to find the median of a data array in linear time. Prove the linear complexity of the algorithm in the worst case. 3 + 4 (CO4) (PO1)

b) What is quasi-polynomial time algorithm? Outline the solution for 0-1 Knapsack problem and justify its complexity as quasi-polynomial. Mention at least one more algorithm that is also quasi-polynomial. 5 (CO1) (PO1)

c) Prove that the complexity of quicksort algorithm is $(nlgn)$. 4 (CO1) (PO1)

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MID SEMESTER EXAMINATION — **SUMMER SEMESTER, 2022-2023**

DURATION: 1 HOUR 30 MINUTES — **FULL MARKS: 75**

CSE 4807: IT Organization and Management

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. It may surprise you that two snowboard enthusiasts who simply wanted a better way to sell their snowboards online have created an e-commerce platform that now has over $275 million in sales and hosts more than 20000 online retailers, including Pixar, Angry Birds, and the Foo Fighters. Tobias Lütke, CEO and founder, has created a business that allows companies of all sizes to set up their own online store, taking a task that used to take months and trimming it down to as little as half an hour. Shopify takes care of everything behind the scenes in return for a subscription fee and transaction fee. The business model focuses on dealing with other businesses. Given the scenario, answer the following questions:

   a) As the CEO of Shopify, what are the skill sets that Tobias Lütke requires? Justify your answer. 5 (CO1) (PO1)

   b) According to Henry Mintzberg, a prominent management researcher at McGill University, a manager has to perform certain roles in an organization. Considering Shopify as an organization which conducts business with other business organizations, what should be the roles of Tobias Lütke in orchestrating such a successful organization as Shopify? 13 (CO1) (PO1)

   c) How many different types of international organizations are there and what type of organization is Shopify? Explain briefly. 4 + 3 (CO1) (PO1)

2. A small pastry shop at a tourist destination wants to expand its business to other locations in a different country. The pastry shop has a loyal customer base and has received positive reviews for its pastries. The owner wants to maintain control over the pastry shop's operations and brand identity while expanding its reach. Regarding the pastry shop, it needs to avoid incurring significant upfront costs because of limited financial resources. At the same time, it desires to keep control over its operations, brand identity, and customer experience. The pastry shop does not have the expertise to manage multiple locations and wants to benefit from the expertise of others.

   a) Mention and justify which type of business operation (importing, exporting, licensing, franchising, strategic alliance) would be the most appropriate for the pastry shop. 8 (CO2) (PO1)

   b) Hofstede discovered some dimensions for helping manager understand different national culture. Analyze them in the context of different countries. 10 (CO1) (PO1)

   c) Management is coordinating work activities so that they are completed efficiently and effectively with and through other people. If you had to choose between being effective or being efficient, which would you say is more important? Justify with examples. 7 (CO2) (PO1)

3. a) PESTEL is an acronym of six factors. What are those factors and why PESTEL analysis is done in strategic management? 10 (CO1) (PO1)

b) What is Organizational Project Management (OPM)? How portfolio, program and project management are integrated here to achieve strategic objectives? Explain with proper diagram. 3 + 4 (CO1) (PO1)

c) What is Cocomo Model? Consider a project size of 300 KLOC is to be developed and the project schedule is not very tight. Identify the project type and calculate the effort and development time considering the developers having very high application experience and very low experience in programming as depicted in Table 1. 2 + 6 (CO3) (PO1)

Table 1: Value of Cost Drivers for Question 3.c

Personal Attributes	Very low	low	Nominal	High	Very High
Analyst Capability	1.46	1.19	1.00	0.86	0.71
Application Experience	1.29	1.13	1.00	0.91	0.82
Software Engineer Capability	1.42	1.17	1.00	0.86	0.70
Virtual Machine Experience	1.21	1.10	1.00	0.90	
Programming language Experience	1.14	1.07	1.00	0.95	

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MID SEMESTER EXAMINATION — **SUMMER SEMESTER, 2022-2023**
DURATION: 1 HOUR 30 MINUTES — **FULL MARKS: 75**

CSE 4839: Internetworking Protocols

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Suppose that computer A is sending a file to computer B using a private Ethernet with no other computers using it. They are connected by a 100m of wire. Bits travel at the rate of $2x10^8 m/s$ in this wire. Suppose the Ethernet has a bandwidth of 10^9 bits per second (gigabit Ethernet). Now, answer the following questions. 3 + 3 + 4 (CO1) (PO2)

   i. Based the on the provided information, what is the latency of the connection? (Assume any missing values.)

   ii. How long would it take for 10^6 bits to finish traveling from computer A to computer B? (Assume any missing values.)

   iii. Suppose you now measure the traveling time in Question 1.a)ii in real network. Furthermore, assume that the computers are fast enough that they do not limit the speed of transmission. Nevertheless, the time you measure is longer than the time you calculated above. What factors could have resulted in this?

   b) Imagine a multinational corporation with offices in major cities worldwide. Employees often travel between these offices for meetings, conferences, and client visits. Despite the implementation of Mobile IP, employees experience significant delays and disruptions in connectivity when transitioning between different network subnets. 5 + 5 (CO1) (PO2)

   What could be the reasons for these delays and disruptions, and what strategies would you implement to address them?

   c) Explain the significance of protocols in ensuring efficient and secure communication across diverse network architectures, and support your discussion with real-world examples. 5 (CO1) (PO1)

2. a) A system uses Reverse Path Forwarding (RPF) algorithm to build multicast trees and deliver multicast packets. There are 100 multicast sources (each generating a single stream of multicast traffic) and 5 groups currently active in the system. What is the number of RPF multicast trees existing in the system? 5 (CO2) (PO2)

   b) A multinational company, X Corp., operates across various geographical regions and wishes to establish separate multicast groups for inter-departmental communication within each region. The company is assigned the Autonomous System (AS) number 8765. The administrator needs to allocate multicast addresses for each regional multicast group. (The multicast address ranges are shown in Table 1). 3 + 3 + 4 (CO2) (PO1)

   Now, answer the following questions.

   i. Identify the range of valid multicast addresses for X Corp. based on its AS number.

   ii. Allocate a unique multicast address for the Marketing department within one of the regions, which is effective for communication within the department. What is its corresponding 48-bit Ethernet address for the LAN using TCP/IP?

   iii. Explain the reasoning behind the selection of multicast addresses, considering the unique communication needs of each department within the multinational company.

Table 1: Multicast Address Ranges for Question 2.b

CIDR	Range	Assignment
224.0.0.0/24	224.0.0.0 224.0.0.255	Local Network Control Block
224.0.1.0/24	224.0.1.0 224.0.1.255	Internetwork Control Block
	224.0.2.0 224.0.255.255	AD HOC Block
224.1.0.0/16	224.1.0.0 224.1.255.255	ST Multicast Group Block
224.2.0.0/16	224.2.0.0 224.2.255.255	SDP/SAP Block
	224.3.0.0 231.255.255.255	Reserved
232.0.0.0/8	232.0.0.0 224.255.255.255	Source Specific Multicast (SSM)
233.0.0.0/8	233.0.0.0 233.255.255.255	GLOP Block
	234.0.0.0 238.255.255.255	Reserved
239.0.0.0/8	239.0.0.0 239.255.255.255	Administratively Scoped Block

c) Consider the network topology and link costs in Figure 1. Assume that a source, **S**, is connected to router **R1** and routers **R5, R7, and R8** have at least one member for the multicast group **G1** connected to them. In the network topology, the MOSPF protocol is used to construct the shortest path tree and to generate the routing table. The source, **S**, sends a multicast message to the multicast group **G1**. 10 (CO2) (PO2)

Draw the multicast routing table for all the routers involved in forwarding the multicast message.

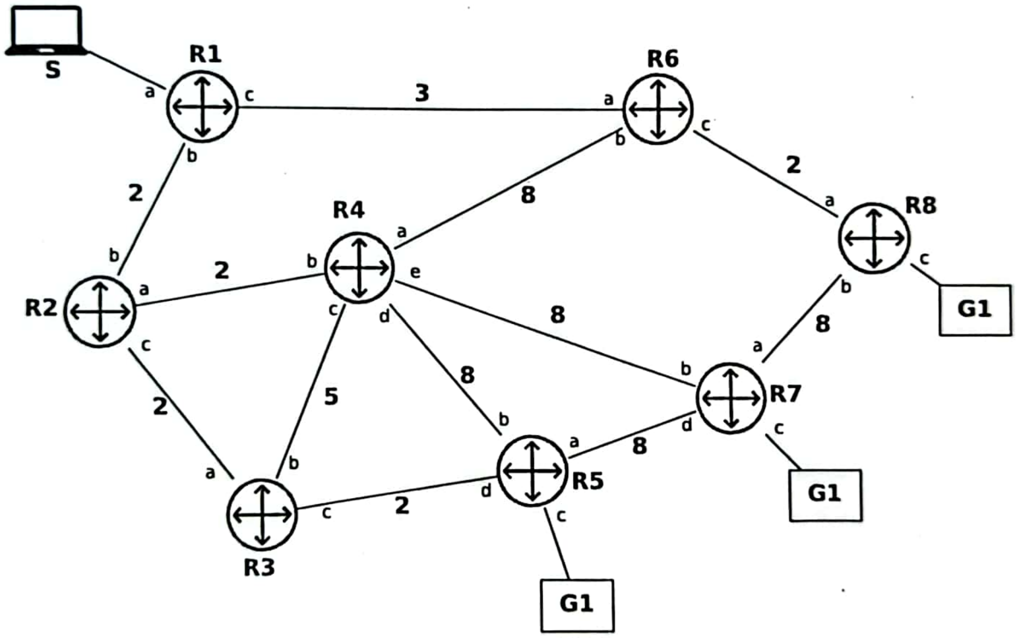

Figure 1: A sample network topology for Question 2.c

3. a) Which algorithm is closer to the path-vector routing algorithm: distance-vector routing or link-state routing? Justify your answer. 5 (CO2) (PO1)

b) Consider the network topology with 11 different Autonomous Systems (ASes) in Figure 2. AS 15, AS 17, and AS 18 are Tier-1 ISPs without any provider, whereas all other ASes have at least one provider. Some BGP sessions between ASes are already shown in the figure, but not all of them. 6 (CO2) (PO2)

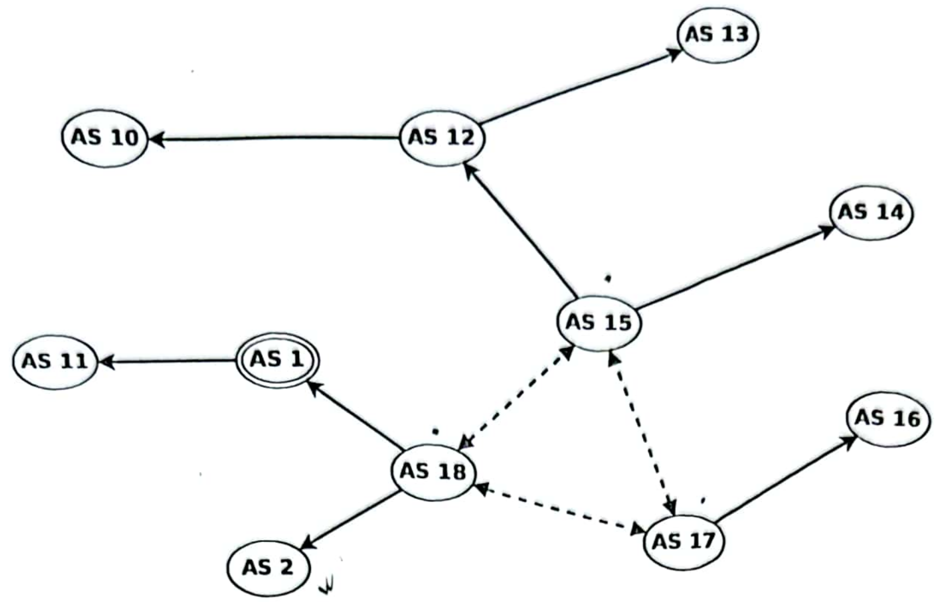

Figure 2: Network topology with a subset of existing BGP sessions for Question 3.b

Single-headed plain arrows point from providers to their customers, while double-headed dashed arrows connect peers. Each AS applies the default selection and exportation BGP policies based on their customers, peers and providers. ASes break ties by preferring the neighbor with the **lowest AS number.**

Each AS XX advertises its prefix XX.0.0.0/24 to all of its neighbors. For instance, AS 12 advertises 12.0.0.0/24. The Tier-1 ISPs (AS 15, 17, and 18) do not advertise any prefix. You are given a complete and sorted list of all incoming BGP messages of AS 1 in Table 2. **"U"** abbreviates a BGP Update message, and **"W"** abbreviates a BGP Withdraw message.

Consider the BGP messages of AS 1 in the table. What might have caused message number 4? Explain the possible causes of receiving the message numbered 4.

Table 2: Stream of BGP messages received by AS 1 for Question 3.b.

AS 1			
#	**type**	**prefix**	**AS path**
1	U	2.0.0.0/24	1 18 2
2	U	10.0.0.0/24	1 11 10
3	U	10.0.0.0/24	1 12 10
4	W	10.0.0.0/24	1 11 10
5	U	11.0.0.0/24	1 11
6	U	12.0.0.0/24	1 12
7	U	13.0.0.0/24	1 12 13
8	U	13.0.0.0/24	1 18 15 12 13
9	U	14.0.0.0/24	1 18 15 14
10	U	16.0.0.0/24	1 18 17 16

c) You are an operator of an AS and you have decided to peer with two new networks, namely AS 195 and AS 466, at router IUTR1. Before establishing the eBGP sessions, you receive the BGP routes that both of them would advertise to you. IUTR1 performs regular BGP path selection process. Further, assume that IUTR1 does not know any additional routing information. For each subtask, you are given a table with three routes, one from each AS (195 and 466), and the currently known route. 3+3+4+4 (CO2) (PO2)

i. Consider the route information in Table 3 for the network prefix 5.21.2.0/24. Find the best route for the destination 5.21.2.35.

Table 3: Route information for the network prefix 5.21.2.0/24

from	prefix	next hop	local pref.	MED	AS Path	IGP cost
current	5.21.2.0/24	12.16.9.1	50	200	260 590	4600
AS 195	5.21.2.0/24	20.79.3.2	100	50	195 439 590	0
AS 466	5.21.2.0/24	13.8.47.25	100	120	466 338 10 590	0

ii. Consider the route information in Table 4 for the network prefix 2.7.8.0/24. Find the best route for the destination 2.7.8.22.

Table 4: Route information for the network prefix 2.7.8.0/24

from	prefix	next hop	local pref.	MED	AS Path	IGP cost
current	2.7.8.0/24	56.22.219.29	150	50	30 89 59 20	0
AS 195	2.7.8.0/24	20.79.3.2	100	100	195 338 89 59 20	0
AS 466	2.7.8.0/24	13.8.47.25	100	80	466 439 20	0

iii. Consider the route information for the network prefix 9.19.2.0/20 in Table 5. There exists no single, most preferred route for 9.19.2.0/20.
Compare the three routes for 9.19.2.0/20 individually with each other. Assume that only two routes are present at the same time. For each pair (i.e. Current vs. AS 195, Current vs. AS 466, and AS 195 vs. AS 466), indicate which route is preferred, and write down the deciding attribute.

Table 5: Route information for the network prefix 9.19.2.0/20

from	prefix	next hop	local pref.	MED	AS Path	IGP cost
current	2.7.8.0/23	56.22.219.29	150	50	30 89 59 20	0
AS 195	2.7.8.0/24	20.79.3.2	100	100	195 338 89 59 20	0
AS 466	2.7.8.0/23	13.8.47.25	100	80	466 439 20	0

iv. Why is a single and preferred route not available for 9.19.2.0/24, and what can you do as a network operator to resolve this issue?